Agent_3

https://gemini.google.com/app/f5004c676c9386f6

User prompt: you are part of a big mission and your task is given in the md file

Response: Step 1: Decoupled Vertical ODE

The time-independent Schrödinger equation for the vertical motion of a particle of mass m under a linear gravitational potential $V(y) = mgy$ for $y \geq 0$ is given by:

$$-\frac{\hbar^2}{2m}\frac{d^2Y(y)}{dy^2} + mgyY(y) = E_yY(y)$$

Rearranging the terms yields:

$$-\frac{\hbar^2}{2m}\frac{d^2Y(y)}{dy^2} + mg\left(y - \frac{E_y}{mg}\right)Y(y) = 0$$

Step 2: Dimensionless Variable Transformation

To eliminate physical dimensions from the ODE, define the dimensionless position variable $z(y)$:

$$z(y) = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3}\left(y - \frac{E_y}{mg}\right)$$

Let $\alpha = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3}$. The derivative of z with respect to y is:

$$\frac{dz}{dy} = \alpha = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3}$$

Applying the chain rule to transform the spatial derivatives:

$$\frac{dY}{dy} = \frac{dY}{dz}\frac{dz}{dy} = \alpha\frac{dY}{dz}$$

$$\frac{d^2Y}{dy^2} = \frac{d}{dy}\left(\alpha\frac{dY}{dz}\right) = \alpha^2\frac{d^2Y}{dz^2} = \left(\frac{2m^2g}{\hbar^2}\right)^{2/3}\frac{d^2Y}{dz^2}$$

Step 3: Conversion to Airy's Canonical Equation

Substituting $\frac{d^2Y}{dy^2}$ and $\left(y - \frac{E_y}{mg}\right) = \frac{z}{\alpha}$ into the rearranged Schrödinger equation:

$$-\frac{\hbar^2}{2m}\left(\frac{2m^2g}{\hbar^2}\right)^{2/3}\frac{d^2Y(z)}{dz^2} + \frac{mg}{\left(\frac{2m^2g}{\hbar^2}\right)^{1/3}}zY(z) = 0$$

Simplifying the coefficients:

1. First term coefficient:

$$-\frac{\hbar^2}{2m}\left(\frac{2m^2g}{\hbar^2}\right)^{2/3} = -\left(\frac{mg^2\hbar^2}{2}\right)^{1/3}$$

2. Second term coefficient:

$$\frac{mg}{\left(\frac{2m^2g}{\hbar^2}\right)^{1/3}} = \left(\frac{m^3g^3\hbar^2}{2m^2g}\right)^{1/3} = \left(\frac{mg^2\hbar^2}{2}\right)^{1/3}$$

Substituting these back into the differential equation gives:

$$-\left(\frac{mg^2\hbar^2}{2}\right)^{1/3}\frac{d^2Y(z)}{dz^2} + \left(\frac{mg^2\hbar^2}{2}\right)^{1/3} zY(z) = 0$$

Dividing through by $-\left(\frac{mg^2\hbar^2}{2}\right)^{1/3}$ yields Airy's canonical equation:

$$\frac{d^2Y(z)}{dz^2} - zY(z) = 0$$

Step 4: Asymptotic Boundary Conditions and Solution Selection

The general solution to Airy's canonical differential equation is a linear combination of the linearly independent Airy functions $\mathrm{Ai}(z)$ and $\mathrm{Bi}(z)$:

$$Y(z) = C_1\mathrm{Ai}(z) + C_2\mathrm{Bi}(z)$$

As $y \to \infty$, $z(y) \to +\infty$. The asymptotic expansions for $z \gg 0$ are:

$$\mathrm{Ai}(z) \sim \frac{1}{2\sqrt{\pi}z^{1/4}}\exp\left(-\frac{2}{3}z^{3/2}\right) \xrightarrow{z\to+\infty} 0$$

$$\mathrm{Bi}(z) \sim \frac{1}{\sqrt{\pi}z^{1/4}}\exp\left(\frac{2}{3}z^{3/2}\right) \xrightarrow{z\to+\infty} \infty$$

For $Y(y)$ to represent a physically bound state, the wave function must be square-integrable and tend to zero as $y \to \infty$:

$$\lim_{y\to\infty} Y(y) = 0$$

Because $\mathrm{Bi}(z)$ diverges exponentially as $z \to +\infty$, physical normalization requires $C_2 = 0$. The physically acceptable vertical solution is therefore:

$$Y(y) = C_y\mathrm{Ai}(z(y))$$

where C_y is a normalization constant and $z(y) = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3}\left(y - \frac{E_y}{mg}\right)$.
